Project Summary:
A Sanctuary in Sound

Emily Amspoker

Background

Interdisciplinary Areas of Study:

- Undergrad: Human-Computer Interaction, Game Design, Design for Learning
- PhD: Human-Centered Computing, Learning Sciences and Technology

Current Focus:

- Qualitative approaches to user research
- Informal learning contexts: how can technology support people's growth and improvement outside of school?



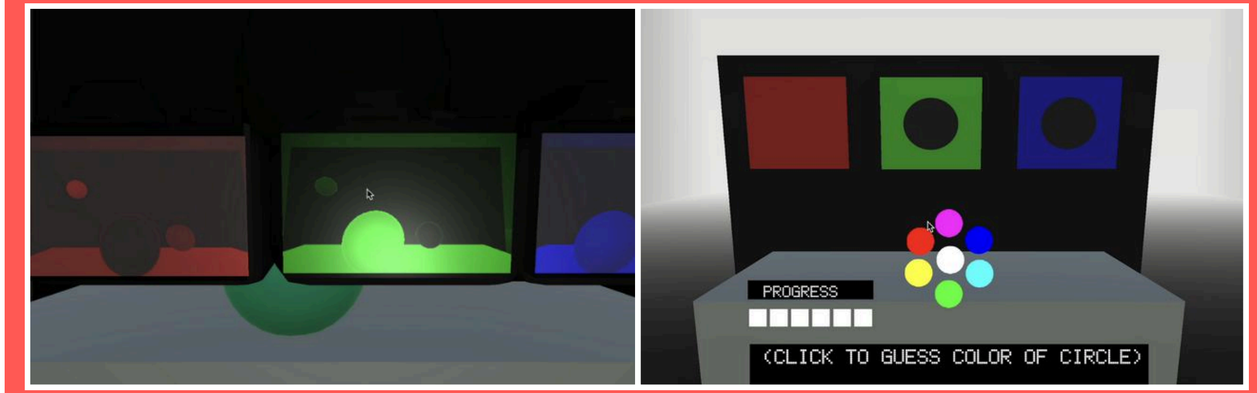
Me at the home of Carnegie Mellon's Integrative Design, Arts, and Technology (IDeATe) program

Past Work & Methods

- **Research Formats:**
  - Educational workshops
  - Playtesting sessions
  - Interviews
  - Large-scale surveys



Project: *Teaching K-12 Learners AI Ethics through a Transformational Game*



Project: *Designing a Digital Experience for Neurodivergent Learners*

PROJECT INTRODUCTION

A Sanctuary in Sound
Project Motivation

- How do we make informal learning spaces like **aquariums and museums** more **accessible** for **blind and low-vision** (BLV) learners?
- How can we teach people about **local ocean data** through **informal learning** interactions?



PROJECT INTRODUCTION

Project Team

Human-Computer Interaction



Georgia Tech College of Computing
School of Interactive Computing

Dr. Jessica Roberts (Interactive Computing)



Emily Amspoker
PhD in HCC Student
(**Project Lead**)



Rhoney Lee
B.S. in CS Student



Solyane Berge
M.S. in CS Student

Sound Design



Dr. Jon Bellona
(Music Technology)

UNIVERSITY OF OREGON

Oceanography



Dr. Amy Bower
(Oceanography)

WOODS HOLE OCEANOGRAPHIC INSTITUTION 1930

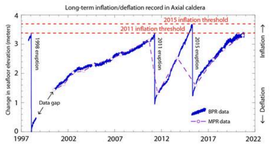


Dr. Leslie Smith
(Oceanography)

Your Ocean Consulting
From the smallest stream to the biggest ocean, your input matters.

Project Context

- Follow-up on previous work creating podcast-style **sonifications** of **simple graphs**
- Initial exploration of **sonifications** of 2D maps using a **touchscreen** (iPad)



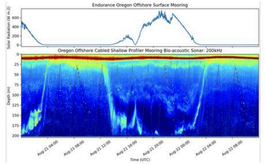
LONGTERM AXIAL SEAMOUNT INFLATION RECORD

Version 2, Full-length Auditory Display

The thirteen tracks make up an auditory display of the Longterm Axial Seamount Inflation Record. The display leverages data from NOAA PMEL that extend the record of the National Science Foundation (NSF) Ocean Observatories Initiative (OOI) data back to 1997. This audio display focuses on the long-term pattern observed by bottom pressure recorders where the seafloor inflates (lifts), then an eruption event occurs, and the seafloor drops.

Listen online to the 7-minute 5-second auditory display (via Samply).

Access or cite the auditory display tracks (via Zenodo).



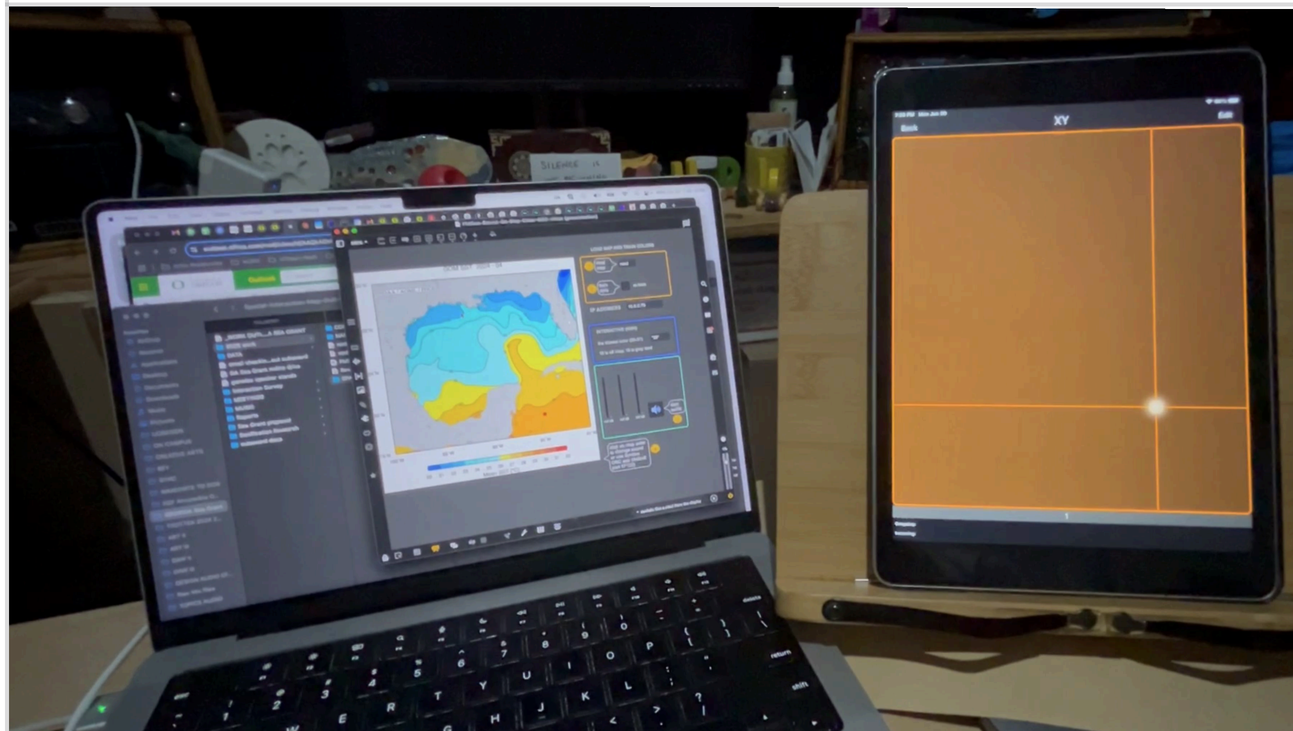
ZOOPLANKTON DAILY VERTICAL MIGRATION GETS ECLIPSED!

Version 2

The nine tracks make up an auditory display of the daily vertical migration of zooplankton off the coast of Oregon. The display is based on the NSF-funded OOI Nugget. Please note that there is no track 1B in this version. We removed track 1B in order to reduce redundancy in the display.

Listen online to the 8-minute 2-second auditory display (via Samply).

Access or cite the auditory display tracks (via Zenodo).



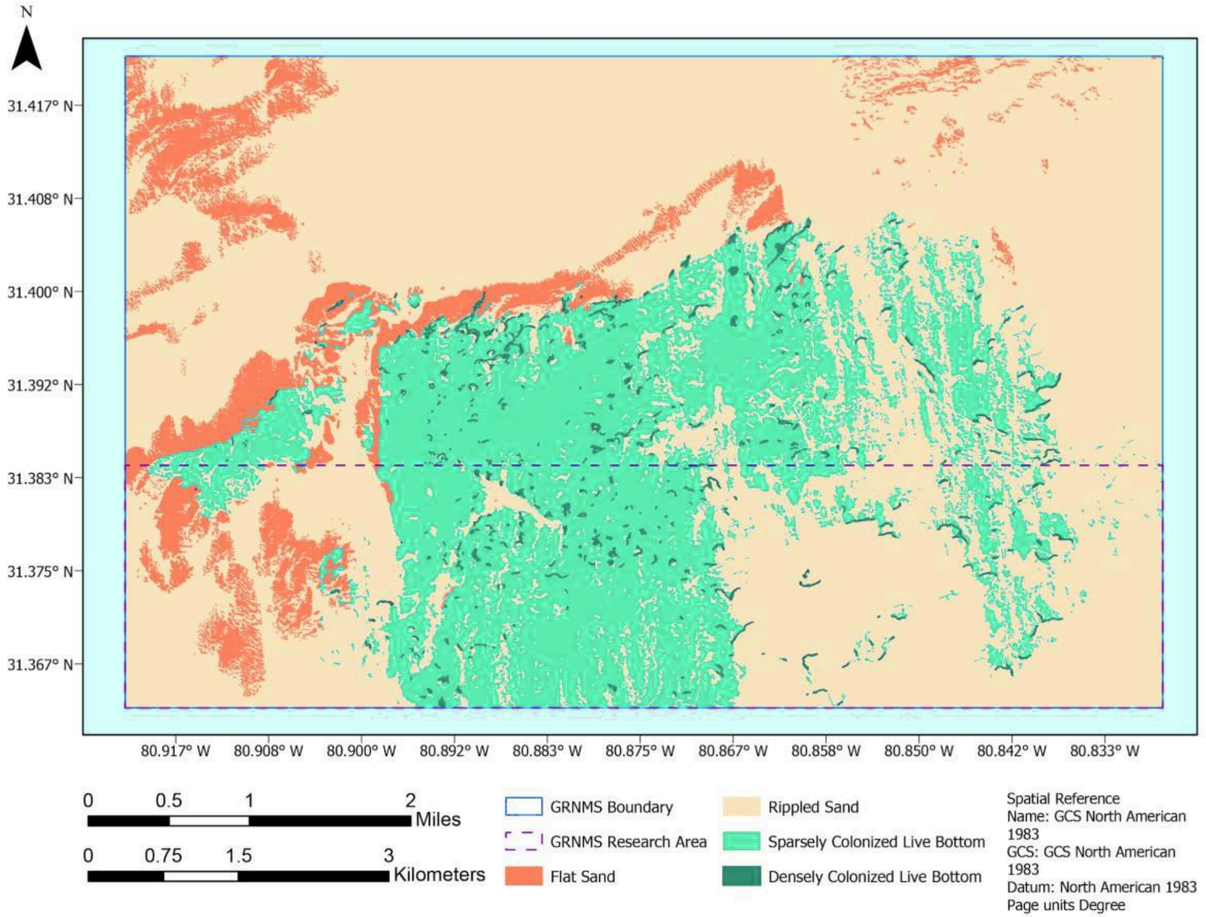
MY ROLE

My Expertise

- Plan to test an aquarium exhibit at the **University of Georgia Aquarium** focused on local marine sanctuaries.
- Needed someone to design and create a prototype of **an interactive display** of **marine habitat data**.



Emily Amspoker
PhD in HCC Student
(**Project Lead**)



Learning Objectives

(LO1) *The ocean floor has **many different types of habitats** in the same area.*

(LO2) *In Gray's Reef National Marine Sanctuary (GRNMS), the **dominant habitat is sand** and the iconic dense reefs are **scattered sporadically** throughout the map.*

(LO3) ***Different animals** are suited to live in different habitat types.*

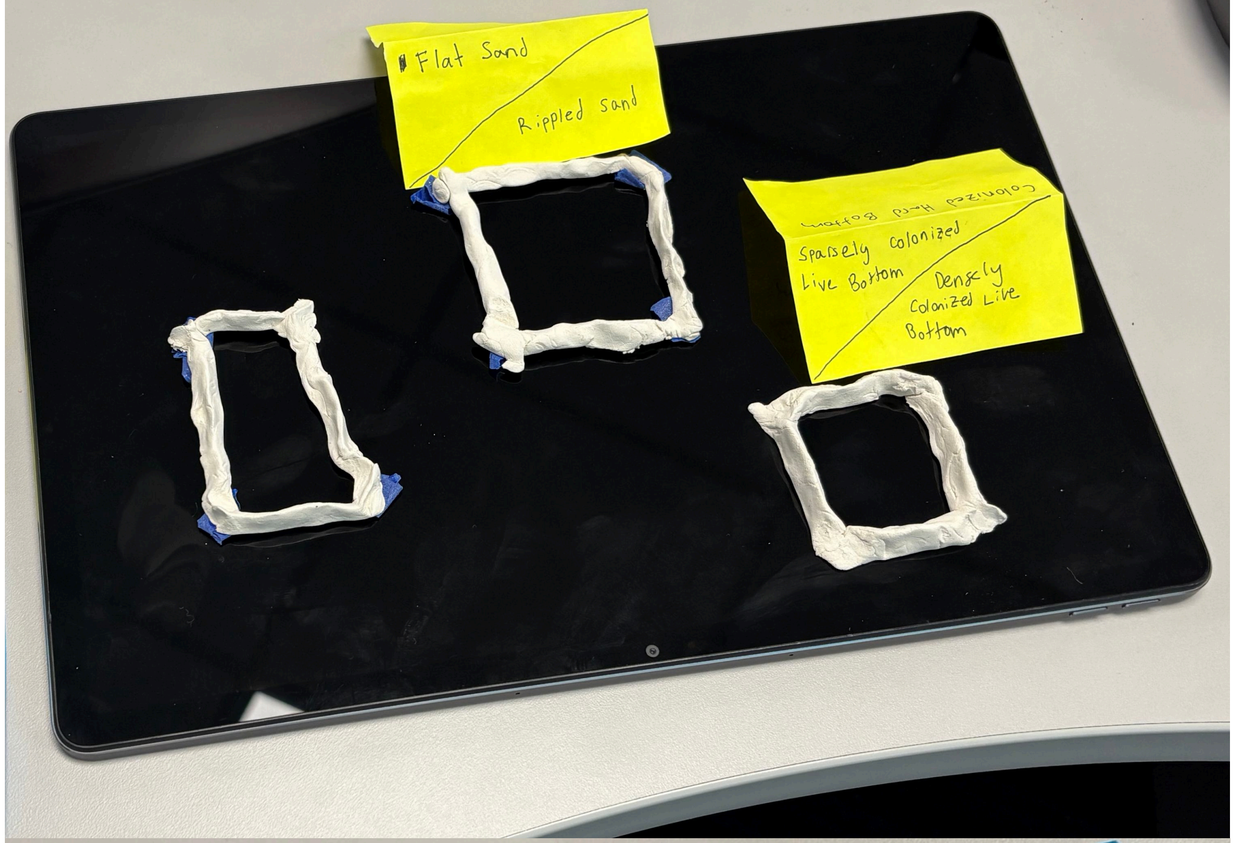
ITERATION #1

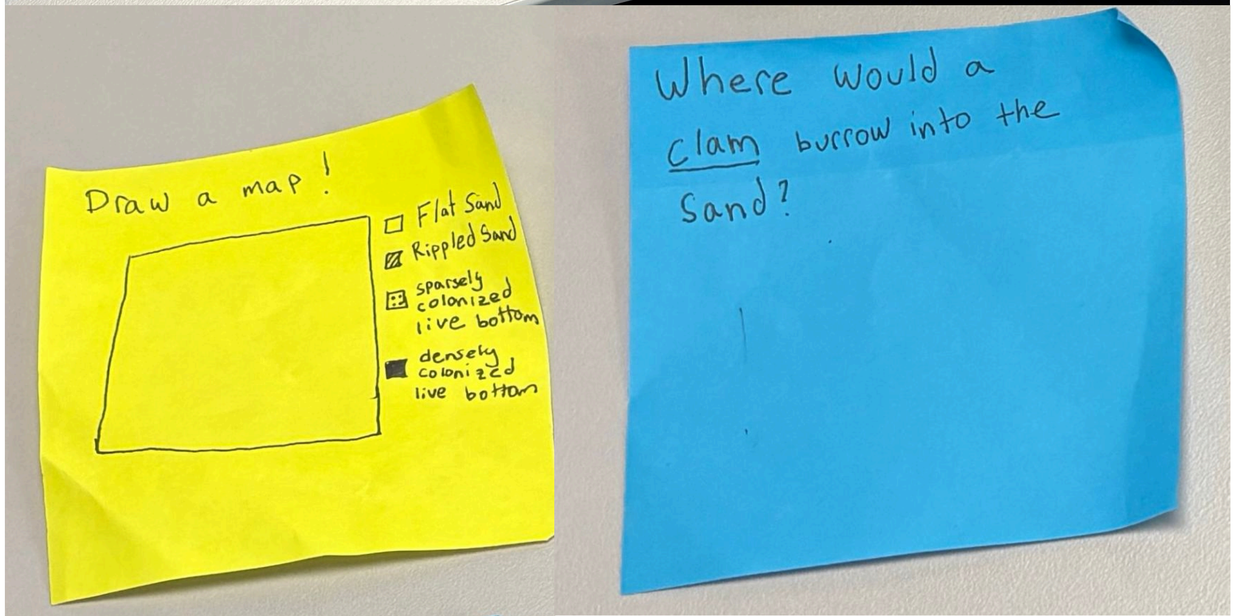
Iteration #1: ‘Stencil’ Prototype

Motivation:
In the previous iPad prototype, free-form exploration of the sound map was **overwhelming and confusing.**

Idea:

- Added **clay barriers** around two touchscreen regions that only had four sounds each.
- Created 2 activities:
  - **Draw the map** you hear.
  - **Find** where a clam would live.





Feedback (Informal Feedback)

- Kept **forgetting** what the sounds meant, confused by the terminology.
- Missed an entire **habitat**.
  - Clay bounding boxes on the map were **too small.**
  - Constrained by the size of the iPad.

> I have no idea what any of these words mean.

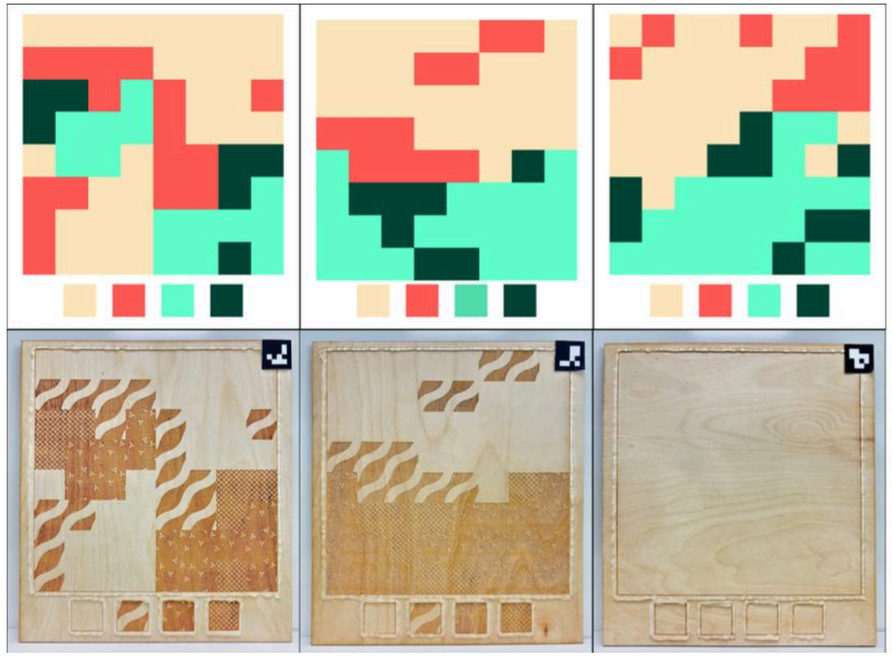
ITERATION #2

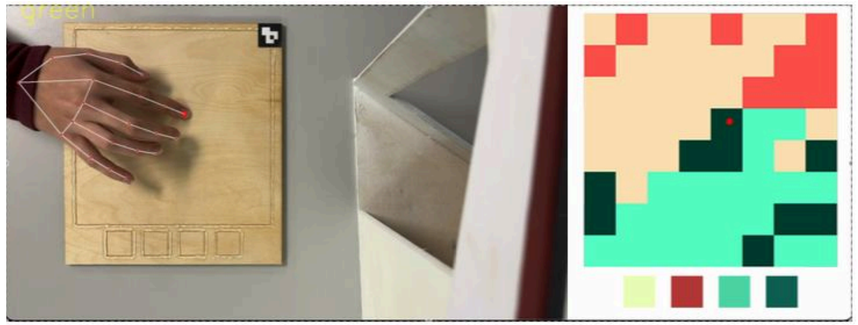
Iteration #2: Computer Vision Prototype

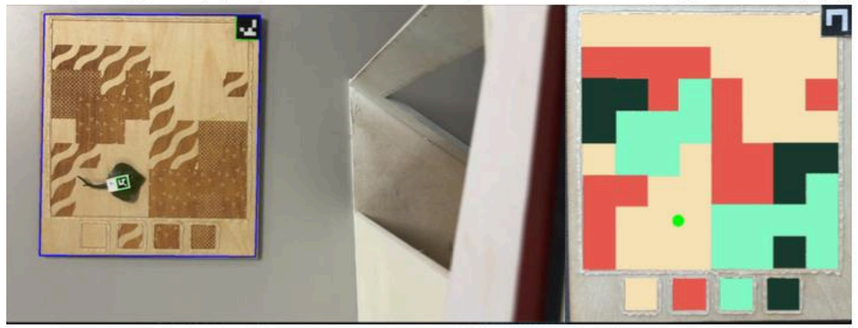
Motivation:
In Iteration #1, the regions of the map were **too small to be detectable**. We wanted to explore texture, but **the tradeoffs and affordances** between texture/sound were unclear.

Idea:

- Switched over to a **computer vision** system.
- Created **three** 8x8 subsets of the map.
  - Texture, texture + sound, sound
- Added a **legend** with auditory descriptions of each map.
- Included detection of **animal figurines**.



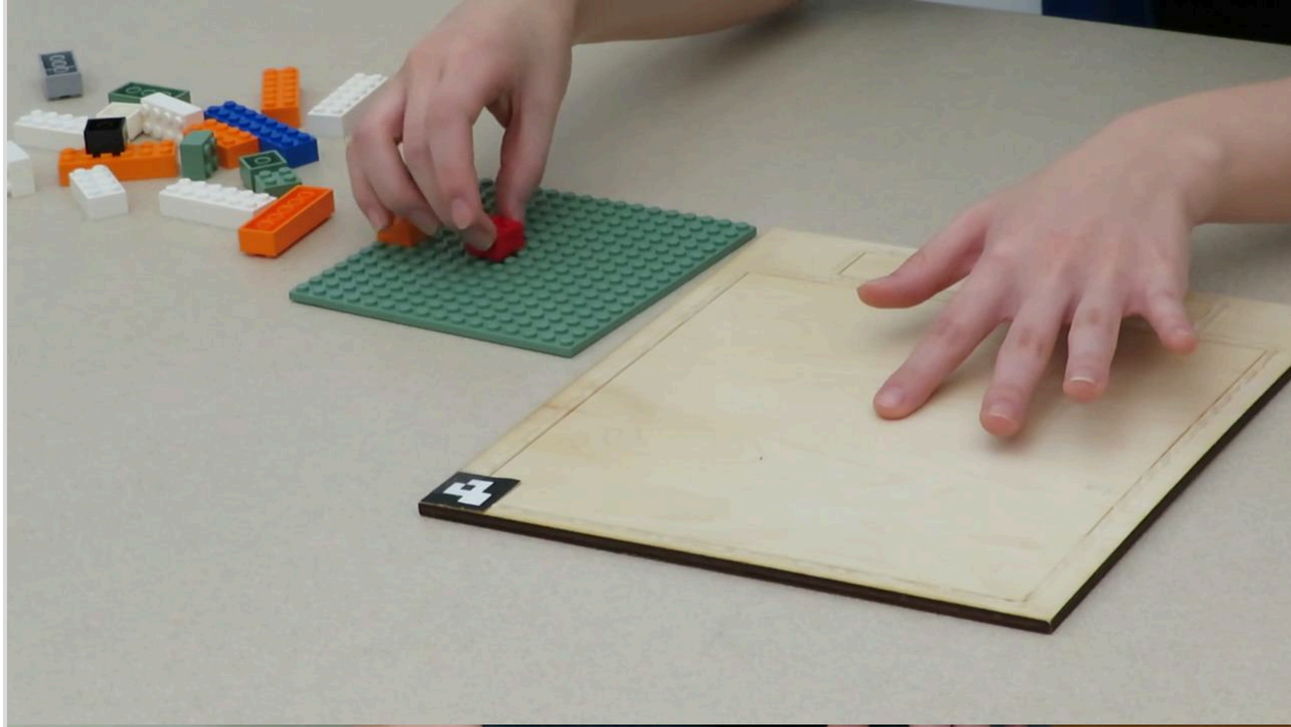


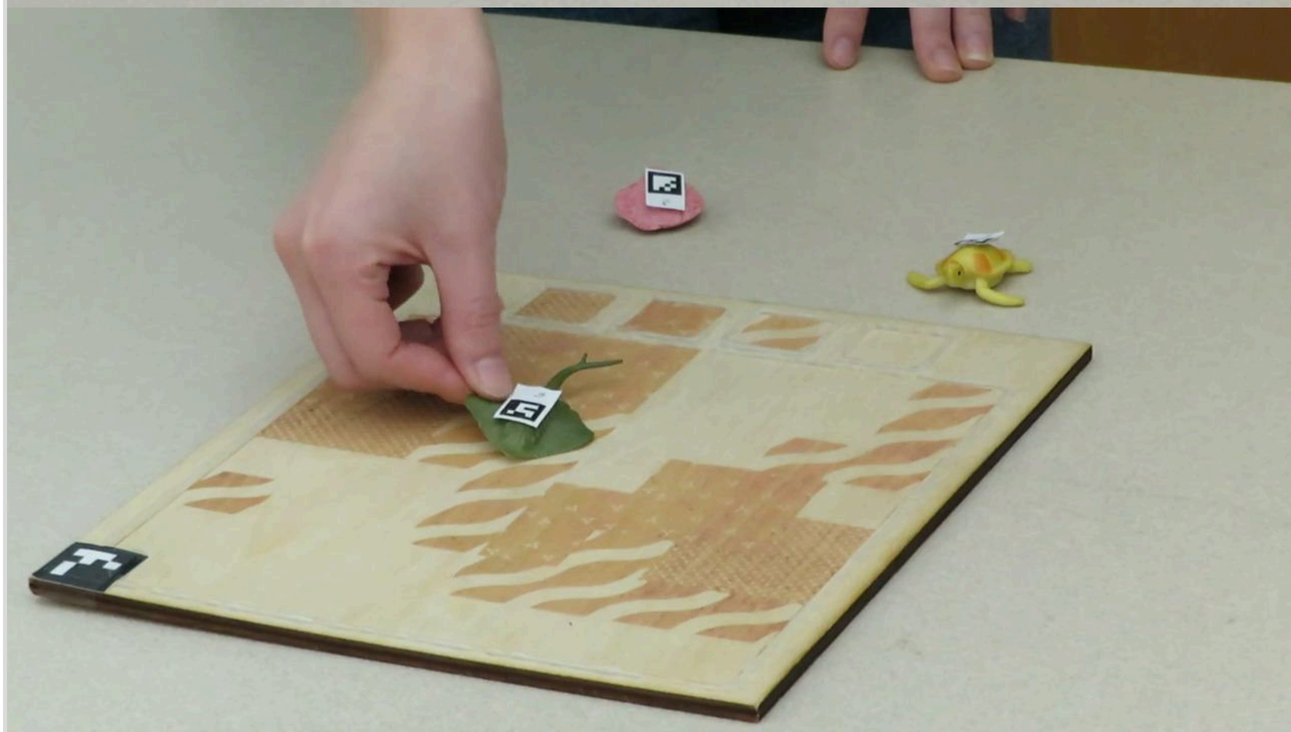


ITERATION #2

Feedback (Semi-Structured Interviews)

- Interviews included:
  - Background questions
  - Map exploration and Lego activity
  - Animal matching activity
  - Post-interview survey
  - Post-interview discussions
- Tested with 4 participants (3 sighted, 1 BLV)
  - Wanted more **diverse** sounds.
  - BLV user counted rows and columns based on their **absolute position on the map**.
  - Sighted users didn’t understand **why they would want to listen to a sound-based map**.





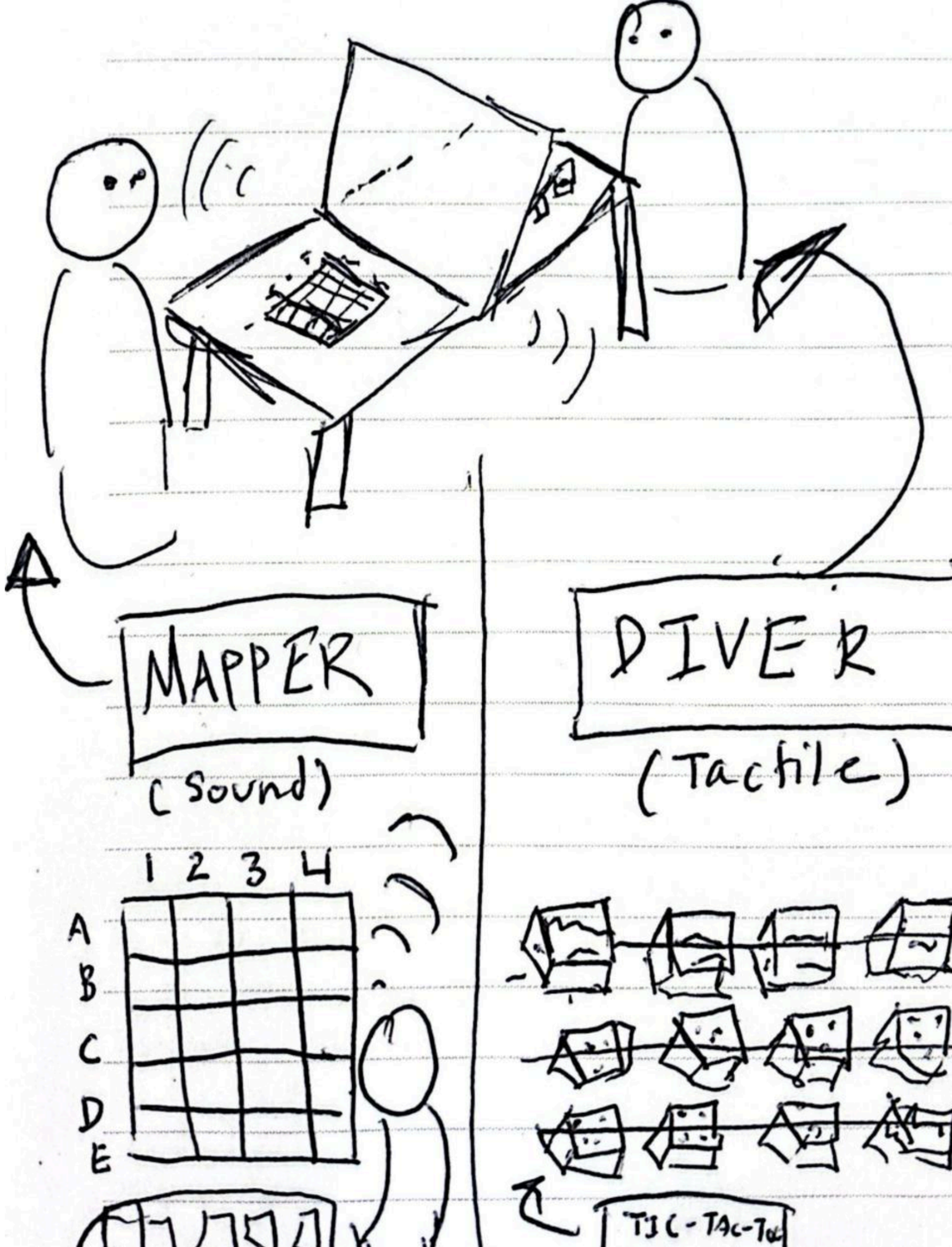
ITERATION #3

Iteration #3: ‘Battleship’ Prototype

Motivation:
In Iteration #2, interviewees wanted **different sounds**. We also wanted to connect the activity to **scientific data collection processes**.

Idea:

- Created a **2-player game** where players work together to map the bottom of the habitat.
  - One player **“drives a boat”** on a sound-only map.
  - The other player
- “Battleship”-style grid system for easy communication about positioning.
- Choice of three kinds of sounds: simple, music, speech



ITERATION #3

Feedback (Aquarium Evaluation)

- Tested with **marine scientists/educators, aquarium visitors.**
- Scientists/educators **identified with the prototype**.
  - “That’s literally us on the boat!”
  - Sparked ideas for new datasets to explore.
- A variety of visitors interacted with the prototype.
  - Asymmetric gameplay helped **families** collaborate.
  - **Speech** was the most effective for the task, but there was **preference for more fun** sounds.





FUTURE WORK

Future of the project

- Creating more robust and permanent displays.
- Designing displays for different kinds of data to work together in a larger exhibit.
- Continuing to evaluate in the context of aquariums and science centers.





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Questions?



https://tiles.cc.gatech.edu/2026/03/06/touchable-sound-tei-2026/